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RESEARCH ARTICLE

Occlusion-Aware Amodal Depth Estimation for Enhancing 3D Reconstruction From a Single Image

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ABSTRACT In various fields, such as robotics navigation, autonomous driving, and augmented reality, the demand for the reconstructing of three-dimensional (3D) scenes from two-dimensional (2D) images captured by a camera is growing. With advancements in deep learning, monocular depth prediction research has gained momentum, leading to the exploration of 3D reconstruction from a single image. While previous studies have attempted to restore occluded regions by training deep networks on high-resolution 3D data or with jointly learned 3D segmentation, achieving perfect restoration of occluded objects remains challenging. Such 3D mesh generation methods often result in unrealistic interactions with graphic objects, limiting their applicability. To address this, this paper introduces an amodal depth estimation approach to enhance the completeness of 3D reconstruction. By utilizing amodal masks that recover occluded regions, the method predicts the depths of obscured areas. Employing an iterative amodal depth estimation framework allows this approach to work even with scenes containing deep occlusions. Incorporating a spatially-adaptive normalization (SPADE) fusion block within the amodal depth estimation model effectively combines amodal mask features and image features to improve the accuracy of depth estimation for occluded regions. The proposed system exhibits superior performance on occluded region depth estimation tasks compared to conventional depth inpainting networks. Unlike models that explicitly rely on multiple RGB or depth images to handle instances of occlusion, the proposed model implicitly extracts amodal depth information from a single image. Consequently, it significantly enhances the quality of 3D reconstruction even when single images serve as input. The code and data used in the paper are available at https://github.com/Seonguke/Occlusion-aware-Amodal-Depth-Estimation-for-Enhancing-3D-Reconstruction-from-a-Single-Image/ for further research and feedback.

INDEX TERMS Occlusion, amodal segmentation, depth estimation, 3D-reconstruction.

I. INTRODUCTION

Currently, technology capable of reconstructing three-dimensional (3D) scenes from two-dimensional (2D) images is crucial in the domains of robot navigation, autonomous driving, and augmented/mixed reality. Traditional 3D reconstruction techniques based on conventional computer vision methods [1], [2], [3], [4], [5] involve the process of matching image keypoints from multiple images taken at various positions to determine the depth information. Through this process, each pixel is mapped to a 3D space, representing 3D information such as point clouds or meshes. With advances in deep learning technology, research on 3D reconstruction methods has gained increased momentum.

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Leveraging multiple sensors to acquire a substantial number of images and depth information, the accuracy of 3D reconstruction can be enhanced through techniques such as multi-sensor fusion [6], [7], [8] and depth information fusion [9], [10], [11]. Furthermore, efforts are underway to address the challenges of using additional sensors or meeting the requirements of multiple image inputs. Estimating depth images from a single image and projecting them into 3D space using the intrinsic parameters of a camera can overcome data acquisition challenges and expand the scope of applications. However, occlusion handling in 3D reconstruction based on a single image is one of the most difficult tasks. In most cases, an image contains multiple objects, often overlapping and resulting in visually occluded areas. Due to insufficient pixel or depth information in these occluded regions, the resulting 3D reconstructions exhibit holes. Such incomplete 3D meshes not only fall short visually but also constrain the natural physical interaction with the graphical objects that are essential in applications of augmented/mixed reality.

This paper proposes an amodal depth estimation technique to enhance the completeness of 3D reconstruction. It takes a single image along with easily extractable modal masks as inputs. By restoring the occluded regions of the modal masks, an amodal mask is generated and utilized for depth estimation, leading to the generation of an amodal depth image. This process is iteratively performed for various depths. In contrast to existing high-performance models that explicitly require multiple RGB or depth images for occlusion handling [12], [13], [14], [15], [16], [17], the proposed model is distinguished in that it implicitly extracts amodal depth information from a single image. The key contributions of this paper are outlined below.

- This is the first study to utilize amodal masks to generate amodal depth images, thereby obtaining the essential multiple depth images required for achieving high-quality 3D reconstruction. Amodal depth images offer crucial geometric information about occluded regions, enabling a significant enhancement in the accuracy of 3D reconstruction based on a single image.
- We propose an iterative approach that repetitively performs amodal-mask generation and amodal-depth estimation processes across different depths. This approach allows for the unrestricted generation of the required number of amodal depth images – even for images with deep occlusion layers.
- To fuse the characteristics of the amodal mask and RGB images effectively, we adopted a spatially-adaptive normalization (SPADE) block. As the transformer blocks of the encoder iterate, the details that tend to fade away are reinforced through the SPADE block, enhancing the representation of the detailed regions within the amodal mask. The output of the SPADE block is forwarded to the decoder responsible for generating the amodal depth, thereby increasing the accuracy of generating depth outcomes for occluded regions.

The rest of this paper is organized as follows. Section II provides an overview of related research. In Section III, the proposed amodal depth estimation method is described, alongside the introduction of the 3D reconstruction system that leverages it. The experimental results are presented in Section IV. The paper is concluded in Section V.

II. RELATED WORK

A. AMODAL SEGMENTATION

Amodal segmentation aims to comprehend the concealed region of a scene, consisting of partially occluded objects, through amodal completion. Completing objects in occluded regions also originates from traditional modal segmentation processes, such as semantic segmentation and instance segmentation. There are three notable learning methods: amodal semantic segmentation, amodal instance segmentation, and self-supervised methods. Amodal semantic segmentation tackles occlusion relationships among objects that share the same semantic class. A notable model in this field is Amodal ERFNet [18]. Amodal ERFNet introduced a joint training approach to predict amodal semantic labels. Given an RGB image as input, each decoder is trained to generate visible semantic labels and amodal semantic labels concurrently. GWSS [19] divides object labels into visible objects and background labels and trains using the Unet architecture. APSNet [20], on the other hand, predicts semantic amodal labels and instance labels in parallel before fusing them with panoptic labels as the final output. Amodal instance segmentation involves completing object instance labels and occluded regions within bounding boxes. One illustrative example is BCNet [21]. BCNet capitalizes on contextual information, with a particular focus on object boundaries, to enhance the instance segmentation quality in scenarios featuring occlusions. The network with this method is a graph convolution network (GCN) [22] that takes into account pixel relationships within occluded regions. The connectivity between objects is reinforced by element-wise addition operations between features used in occluders and those used in predicting occludees. Kar et al. [23] proposed a method using convolutional neural networks (CNNs) to predict bounding boxes that consider the object's size, leading to the first segmentation of amodal instances. ORCNN [24] extends Mask R-CNN by adding a segmentation branch to perform predictions for both amodal and modal masks. The self-supervised learning technique employs available modal segmentation ground truth (GT) labels to model object occlusions and produce amodal masks in situations where creating GT labels for amodal masks is challenging. Within PCNet [25], occlusion relationships are established using instance labels to construct an order graph based on mask areas between objects and adjacent points. Following this, a partial convolution-based encoder-decoder network is utilized to learn amodal masks through self-supervised

learning. In cases where an object remains uncovered by other objects, it is output as is. However, in occluded situations, a reconstructed amodal mask is generated. Amodal-VAE [26] proposes predicting visible masks and complete masks in a trained decoder without amodal annotations. ASBU [27] enhances PCNet's performance by predicting the uncertainty of the boundaries of amodal objects.

B. SINGLE IMAGE DEPTH ESTIMATION

Single-image depth estimation employs two main approaches: metric depth estimation [28], [29], [30] and relative depth estimation [31], [32], [33]. Metric depth estimation aims to predict depths in physical units and is trained on specific target datasets. It is often associated with poor generalization and a high risk of overfitting. LeRes [28] utilized a 3D network to predict shifts and focal lengths, correcting the depth to create accurate 3D point clouds. Depth map decomposition [29] estimates precise meter-scale depth maps by fusing gradient maps and normalized depth maps from a shared encoder and three decoders. DS-SIDENet [30] makes single-image depth predictions by applying the depth map range predicted in the first stage to the second stage's predictions. The relative depth estimation method uses the scale-shift invariant loss for training on normalized diverse datasets, enabling generalization and creating relative depth distributions between objects. Midas [31] introduced a method that allows learning from various scenes, datasets, and even sources, such as 3D movies, promoting model generalization across diverse domains. It employs scale-invariant loss and outputs a semantically distinguishable depth for any input image. DPT [32] effectively enhances relative depth performance outcomes using a vision transformer (ViT) [34] encoder structure and token fusion through reassembly. BoostingMonocularDepth [33] proposes double estimation and a merging network to generate detailed relative depths that represent object intricacies effectively.

C. 3D RECONSTRUCTION FROM A SINGLE IMAGE

3D reconstruction is a technology that generates 3D models from 2D images [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [35], [36], [37], [38]. In the past, structured lighting approaches were widely used, involving 3D scanners and multiple images. However, with recent advancements in deep learning, it has become possible to reconstruct 3D from a single image. Various methods have been studied to predict 3D occupancy using deep learning networks trained on single-view images and depth maps. These methods produce high-resolution 3D models and eliminate the need for additional 3D scanners, resulting in cost savings. Among the techniques for single-image-based 3D reconstruction, SSCNet [36] projects 3D space information onto 2D images as input. Then, using a CNN, it processes the 2D image and converts it into a truncated signed distance function (TSDF) format that corresponds to the 3D space. This approach utilizes a 3D-convolution-based encoder-decoder architecture to predict semantic labels and occupancy for the voxels. By jointly learning segmentation labels and geometry information, sparse inputs can be densified. Panoptic 3D reconstruction [37] takes RGB images as input and predicts depth and instances using a 2D network, incorporating a weight-sharing technique. Subsequently, the data is projected into 3D space, where the 3D network utilizes input features and depth information to embed 3D coordinates and corresponding features into a TSDF representation. This is done to predict the occupancy, instance, and semantic labels. Finally, the instance and semantic labels are used to generate panoptic labels through post-processing. By harmonizing the segmentation labels with the generated occupancy based on the projected points, the 3D geometry results are enhanced. However, when using projected points based on depth captured from a single image, the information for occluded regions becomes ambiguous, leading to suboptimal 3D geometry performance in occluded areas.

D. DEPTH IMAGE INPAINTING

Depth image inpainting [12], [13], [14], [15], [16], [17] aims to estimate missing or corrupted depth information in depth maps or 3D point clouds based on the surrounding depth information and context. Various algorithms, including image inpainting techniques, geometric priors, and deep learning approaches, are utilized to achieve this goal. Inpainted depth maps can be used in conjunction with RGB images or other visual information for various depth-based applications, such as the reconstruction of 3D scenes, virtual object placement, depth-based rendering, or depth-aware image editing. Related studies include those focusing on 3D-Photo [17] and VeI [16]. In 3D-Photo, an edge inpainting network is employed to inpaint the depth of the composited region. Subsequently, the inpainted depth edges are connected with contextual colors, and a color inpainting network is applied to generate the inpainted color. Similarly, by applying an inpainting network to the inpainted depth edges along with the contextual depth and the input, the inpainted depth is generated. VeI [16] uses randomly generated boxes as input for depth images. The network generates depth and normal images as outputs, which are then utilized as inputs for surface attention and dilated convolution encoders. The resulting feature maps serve as inputs for the decoder, ultimately producing a depth image with restored mask regions. While there is active research in the form of depth-map-related studies focusing on topics such as depth image inpainting, such research is relatively scant compared to that focusing on RGB images.

III. PROPOSED METHOD

A. OVERVIEW

Figure 1 presents the overall structure of the proposed 3D reconstruction system. It is divided into two main stages: iterative depth predictions for occluded regions and 3D reconstruction using these predictions. The input for the first stage consists of a single RGB image, denoted as I_0,

and N instance segmentation masks ($M_{N-1}, M_{N-2}, \ldots, M_0$) that predefine occlusion relationships. These masks can be easily obtained through instance segmentation techniques. N represents the number of depth levels to be distinguished, and it can be determined either initially or based on the instance segmentation results. The iterative depth estimation process is repeated N times. At the initial iteration $t = 0$, conventional monocular depth estimation is applied to input image I_0, yielding an the amodal depth aD_0. Subsequent iterations utilize the proposed amodal mask generation (AMG) and amodal depth estimation (ADE) methods to generate amodal depths. For AMG, the occluder mask aM_{t-1} and the occludee mask M_t are provided as inputs. Referring to aM_{t-1}, the occluded regions of M_t are restored, resulting in output aM_t, which becomes the occluder mask for the next iteration. ADE takes the restored occludee mask aM_t from AMG and the input image with the occluder removed, denoted as I_t. In Figure 1, the process depicts the prediction of the amodal depth aD_1 for a sofa area occluded by a table, which is the occluder. This process is repeated iteratively from near to far depths, generating $aD_0, aD_1, \ldots, aD_{N-1}$ sequentially. Various techniques, such as depth inpainting [12], [13], [14], [15], [16], [17], have been proposed to fill occluded regions in depth images. However, these methods are better suited for small mask areas or simple distribution patterns. Dealing with larger-scale mask regions, especially those involving complex structures such as furniture or substantial occlusion areas, the use of generative models such as inpainting to restore depth all at once is challenging. In this paper, we address this issue by breaking it down into two sub-problems. AMG provides a solution to the simplified binary classification problem of finding masks in occluded regions. The information obtained from these masks serves as crucial prior knowledge about the complete object appearance, thereby alleviating the difficulty of depth generation in ADE.

In the second stage, we employ the SG3N model [37], which is specifically designed to tackle not just 3D reconstruction, but also semantic and instance segmentation from monocular images. However, as this paper focuses on 3D reconstruction, we have removed the segmentation modules from SG3N and utilized only the modules related to scene geometry restoration. In the original SG3N model, depth extracted from the RGB image serves as a prior. The model then lifts the 2D features into a 3D volume using the known camera intrinsic and the predicted depth estimates. The depth image is back-projected into a 256^3 volumetric grid and encoded as a TSDF along the viewing direction, where positive values are in front of the predicted surface, zero encodes the surface itself, and negative values represent unseen regions. In the second stage of our model, instead of a single image, N depth images are projected into 3D space, which includes information about occluded regions. The 3D scene reconstruction is then predicted from the feature volume using a sparse generative approach. SG3N is trained to transform sparse point clouds into dense ones, takes P^{Sparse} as input, and produces a dense point cloud, P^{Dense}. Designed as an encoder-decoder network, SG3N includes two sparse convolutions, θ_S [39], and three 3D dense convolutions, θ_D, predicting the 3D geometry in a coarse-to-fine manner. The final output is a mesh generated from P^{Dense} using the marching-cubes algorithm [40].

B. AMODAL MASK GENERATION

Figure 2 illustrates the process of AMG. The network adopts an encoder-decoder architecture based on partial convolutions proposed in PCNet [25], as shown in Figure 2a. It takes the restored occluder mask from the previous iteration, aM_{t-1}, and the occludee mask of the current iteration, M_t, as inputs, and generates an amodal mask aM_t with accurately restored occluded regions from the occludee mask. However, since there is no GT available for aM_t, AMG cannot be trained using a supervised learning approach. Therefore, as depicted in Figure 2b, it is trained in a self-supervised manner. An arbitrary mask M in the first column serves as the GT for AMG. A randomly placed gray occluder occludes an arbitrary mask M, as shown in the second column. The image in the third column is obtained by subtracting the occluder from M, and this can be restored through AMG, as seen in the fourth column. This restored mask is compared with the GT, M to train AMG. Here, the gray occluder serves as aM_{t-1} in Figure 2a, the image with the occluder removed from M represents M_t in Figure 2a, and the image in the last column restored through AMG corresponds to aM_t. To prevent over-completion in cases where there is no occlusion, training is divided into two cases: Case 1 involves occlusion, whereas Case 2 involves no occlusion.

C. AMODAL DEPTH ESTIMATION

ADE aims to predict the depth of the scene beyond the occluders, that is, the areas hidden by occluders. While color information in the image generally plays a crucial role in depth estimation, it must be strictly excluded when undertaking amodal depth estimations. Therefore, as illustrated in Figure 3, the features of the occluder (in this case, the table) are explicitly removed from the input I_t to prevent them from being used for amodal depth inference. However, these erased regions retain ambiguity. The sofa in I_t does not exhibit a complete shape, and its boundary with the back wall is not well defined. In such cases, directly predicting the amodal depth beyond the occluder table is a challenging problem. Thus, an amodal mask that restores the information of the sofa corresponding to the erased region through binary classification in AMG is provided as input to ADE. By explicitly indicating the area where the relevant features for estimating the depth must be referenced, the challenge of predicting depth in occluded regions can be addressed.

Figure 3a shows the structure of the network for ADE. It consists of a transformer-based encoder, a SPADE

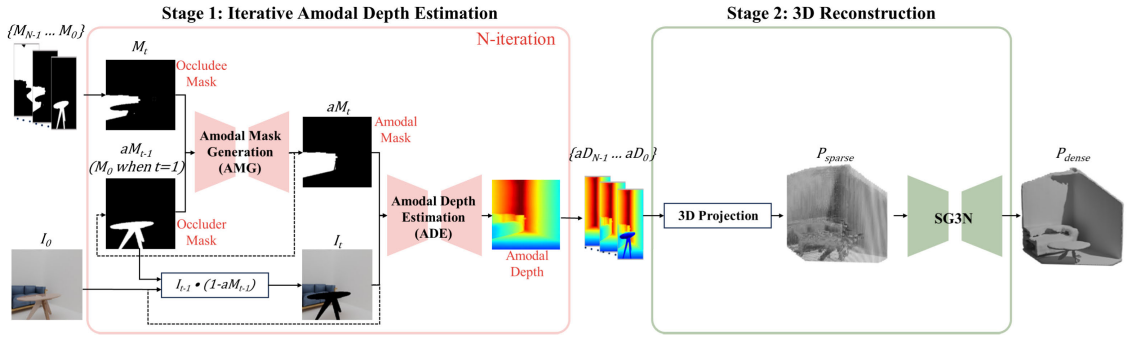


FIGURE 1. Overall structure of the proposed 3D reconstruction system. It is divided into two main stages: iterative depth predictions for occluded regions and 3D reconstruction using these predictions. The iterative depth estimation process is repeated N times. At the initial iteration (t = 0), conventional monocular depth estimation is applied to the input image. Subsequent iterations utilize the proposed amodal mask generation (AMG) and amodal depth estimation (ADE) methods to generate amodal depths. These amodal depths are provided to the 3D reconstruction networks to supply rich occlusion information.

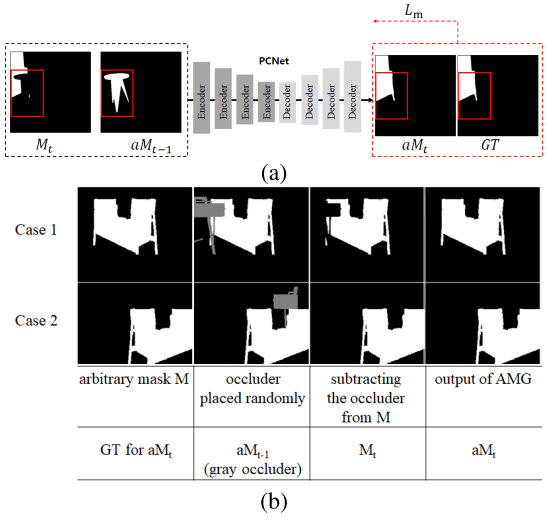


FIGURE 2. The process of Amodel mask generation (AMG) (a) The network structure of AMG adopts an encoder-decoder architecture based on partial convolutions proposed in PCNet [25] (b) AMG is trained in a self-supervised manner.

fusion block, and a decoder. The transformer block used is a variant of a vision transformer [34], specifically vit_base_resnet50_384. Inputs I_t and aM_t undergo patch embedding in 16 × 16 units. The encoder involves four transformer blocks and down-sampling stages, where the output tokens from each stage are used as inputs for the next stage. By providing aM_t as an input to the transformer blocks, it helps indicate the area containing features relevant to the occluded region. However, due to the normalization layers within the transformer blocks, there may be a loss of detail in aM_t, such as in the case of the sofa legs. Therefore, to enable spatially-adaptive normalization, we incorporated a modified version of the well-known SPADE [41] operation, adapted to the purpose of ADE, to additionally provide segmentation information. Here, we aimed to strengthen the role of the amodal mask in depth estimation by providing an amodal mask of the same size as the output feature map of the transformer block. This technique is applied to ADE to enhance the fine details of a_t. The SPADE fusion block exists between the encoder and decoder at each stage. The output tokens of the transformer block are reassembled into a 2D-image-shaped F_{stg_n}, and aM_t is resized to match the size of F_{stg_n}. These features are then inputted to the SPADE fusion block of each stage. Figure 3b provides a detailed view of the internal structure of the SPADE fusion block. aM_t is transformed into an activation map using convolution and rectified linear unit (ReLU) activation. Next, convolution is applied to this activation map, resulting in an element-wise multiplication operation with F_{stg_t}, followed by the application of convolution to the activation map for element-wise summation. The output of the SPADE fusion block is passed to the decoder. The feature map that has been upsampled by a factor of 2 in the previous stage's decoder block and the output of the SPADE fusion block are concatenated along the channel dimension. After passing through the decoder block four times, the amodal depth map aD_t is produced.

D. ITERATIVE DEPTH ESTIMATION

Figure 4 illustrates the sequential process of estimating aD_0 to aD_{N-1}. This example is shown for N=3. In the initial iteration $t = 0$, conventional mono-depth estimation is performed to generate aD_0. Starting from iteration $t = 1$, AMG and ADE collaborate. AMG takes M_0 and M_1 as inputs. The ambiguity of occludee mask M_1 is resolved with reference to occluder mask M_0, resulting in a complete amodal mask aM_1. In ADE, amodal mask aM_1 and I_1 are taken as inputs to generate aD_1. Here, I_1 is an image obtained by removing the occluder M_0 region from the image I_0 used in iteration $t = 0$. By explicitly removing occluder information and providing the locations of neighboring pixels to be referenced from the amodal mask, it becomes possible effectively to infer and fill the depths of large occluded areas. This goes beyond the limitations of conventional depth inpainting techniques that address small masked areas. The process for the next iteration, $t = 2$,

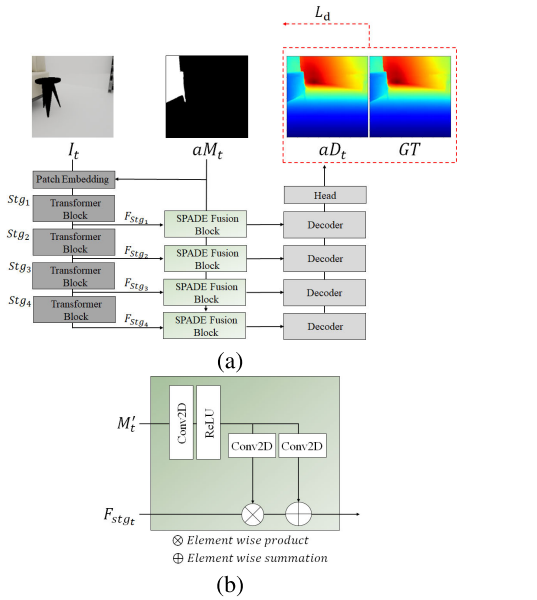


FIGURE 3. Amodal depth estimation (ADE) predicts the depth of the scene beyond occluders. (a) The network structure of ADE, which consists of a transformer-based encoder, a SPADE fusion block, and a decoder. (b) The SPADE fusion block additionally provides segmentation information. It enhances the role of the amodal mask in depth estimation by providing an amodal mask of the same size as the output feature map of the transformer block.

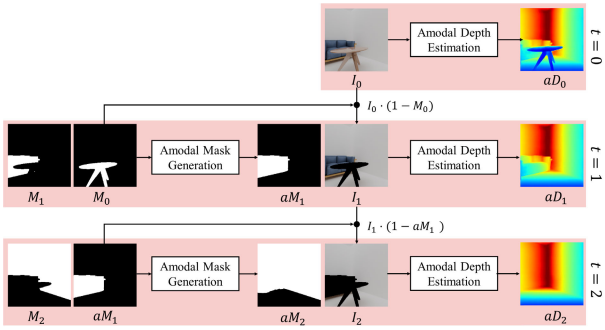


FIGURE 4. The process of iterative depth estimation. This example is shown for N=3. In the initial iteration t = 0, conventional mono-depth estimation is performed. Starting from iteration t = 1, AMG and ADE collaborate.

is similar with iteration $t = 1$, where the amodal mask aM_1 generated in iteration $t = 1$ becomes the occluder for AMG in iteration $t = 2$. This is utilized to fill in the ambiguity of the empty regions of occludee M_2, yielding the amodal mask aM_2. In ADE, aM_2 and I_2 are used to predict aD_2.

E. LOSS

In the proposed system shown in Figure 1, three deep learning networks are utilized. The loss function L_m for AMG is defined as (1). The binary cross-entropy (BCE) loss L_{BCE}, commonly used in instance segmentation, measures the discrepancy between the predicted probability distribution and the actual labels to train the model. In the AMG network, we assigned different weights to the occluded and non-occluded regions in the predicted mask pixels. L_{BCE}^{occ} corresponds to the area of $aM_t \cdot aM_{t-1}$, representing the occluded regions, while L_{BCE}^{nocc} corresponds to $aMt \cdot (1 - aM_{t-1})$, representing the non-occluded regions. The values of α and β are set to 5 and 1, respectively, to prioritize the restoration of occlusion areas.

$$L_m = \alpha \cdot L_{BCE}^{occ} + \beta \cdot L_{BCE}^{nocc} \tag{1}$$

For the training of the ADE, L_{ssi} [31] and $\log(L_{pixel})$ [42] are utilized. L_{ssi} is employed to learn invariant scale and shift for the predicted depth, which is applied to mean square error (MSE). This helps maintain the context of depth information, enabling the preservation of context in occlusion areas of the image. $\log(L_{pixel})$ represents the MSE in the log space, calculated for each pixel by the difference between the predicted and actual values. Applying a log scale to the error for each pixel aids in reducing the magnitude of errors. This is used to learn metric depth considering the magnitude and shift of relative depths. The loss function L_d for ADE, which combines L_{ssi} and $\log(L_{pixel})$, is provided in (2).

$$L_d = L_{ssi} + log(L_{pixel}) \tag{2}$$

The loss function L_g for SG3N is defined as (3). L_{l1} measures the distance values of the predicted surface geometry, whereas L_{BCE} is used for the geometry occupancy representing the voxel occupancy ratio.

$$L_g = L_{l1} + L_{BCE} \tag{3}$$

IV. EXPERIMENTS

A. ENVIRONMENT

1) DATASET

Existing datasets for learning 3D reconstruction from a single image, such as [43], [44], do not provide the amodal depth as the GT. Therefore, based on the public dataset 3D-FRONT [43], we created the Amodal-3D-FRONT dataset using the rendering tools Blenderproc [45] and SDF-gen [46]. The 3D-FRONT dataset consists mainly of synthesized indoor images, featuring 18,968 rooms adorned with a variety of 3D furniture objects. The dataset includes a furniture subset of 13,151 items, accompanied by high-quality textures. To generate the 2D images to be used as the GT, we employed Blenderproc [45] to render the 3D-FRONT dataset. Views of size 320 × 240 were rendered in the 3D rooms. By randomly sampling combinations of rooms and 3D objects, diverse views with occlusions were generated. To capture each scene, objects were fixed at a height of 0.75m above the floor and rotated 90 degrees around the x-axis, with this followed by setting random positions and rotation angles. To obtain amodal images, three scenes with different depths were captured sequentially by successively removing objects from the scene. Cases with excessively large or small occlusions were excluded. To generate the 3D geometry for GT, SDFGen is used to create TSDF representations. Leveraging the camera pose information from 3D-FRONT and Blenderproc's rendered views, we composed all object

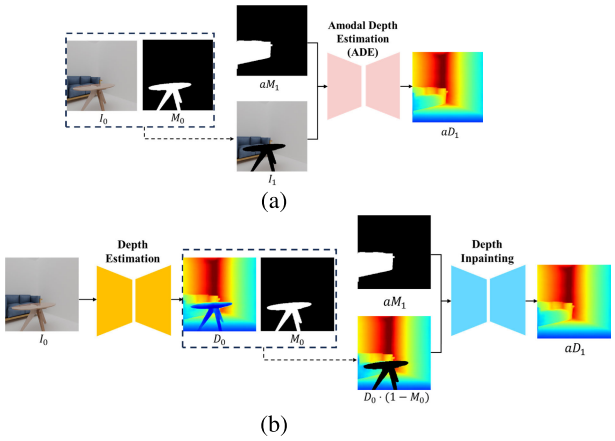


FIGURE 5. Comparison between depth inpainting techniques and the proposed ADE method. (a) The proposed ADE method takes amodal masks and masked RGB images as inputs and generates the depth using a single network. (b) Typical depth inpainting techniques take amodal masks and masked depth images as inputs, requiring an additional estimation process for the input depth.

information within the camera frustum into a TSDF format 256^3 voxels in size. Through this process, a single set of data for a scene was generated, consisting of three RGB images, three depth images, instance labels, and the TSDF representation of the scene. Our Amodal-3D-FRONT dataset comprises approximately 9,000 sets. Among them, 6,000 sets were used for training, and 3,000 sets were used for testing.

2) TRAINING DETAILS

The training was conducted using the PyTorch framework on an RTX 3090 Ti GPU. For AMG training, a learning rate of 0.001 was set, and a stochastic gradient descent optimizer was employed. Among a total of 560,000 iterations, the learning rate was multiplied by 0.1 at the 32,000th and 48,000th iterations to decrease it. For ADE training, an initial learning rate of 0.001 and a batch size of 2 were utilized. During 150,000 iterations, the learning rate was reduced by a factor of 0.1 at the 20,000th and 40,000th iterations. For SG3N training, the batch size was fixed at 1 and training was conducted for the decoder. Over the course of 100,000 iterations, the model was initially trained to output resolutions of 64^3 for the first 3,000 iterations, followed by 128^3 resolutions for the next 6,000 iterations. Subsequently, the model was trained to output 256^3 resolutions starting with the 9,000th iteration for all subsequent decoder outputs. The Adam optimizer was employed with an initial learning rate of 0.001, and the learning rate was multiplied by 0.1 at the 25,000th and 35,000th iterations.

3) EVALUATION METRICS ON AMODAL DEPTH

Occlusion is one of the ill-posed problems in computer vision. Especially for 3D reconstruction based on depth information which is easily extractable from monocular images, an assumption can be made that the presence or absence of depth information for occluded regions would significantly impact the reconstruction results. To validate this hypothesis, we attempted mesh reconstruction using both GT depth and GT amodal depth, as illustrated in Figure 6. As evident from this figure, utilizing GT amodal depth effectively restores occluded areas such as the sofa area obscured by the table. From these preliminary experiments, we indirectly confirmed the significance of amodal depth as a means to enhance the quality of 3D reconstruction. Consequently, we concluded that it is necessary to evaluate not only the quality of the final mesh but also the quality of amodal depth.

To evaluate the performance of amodal depth estimation, the quality of the output aD_1 at iteration $t = 1$ was measured. For a performance comparison, VeI [16] and 3D-Photo [17], which are depth inpainting techniques, were used. The PCNet-C_D and TFill$_D$ models are versions retrained using depth data based on PCNet-C [25] and TFill [47], which were originally designed for RGB inpainting. The comparison between depth inpainting techniques and the proposed ADE was conducted using the methodology shown in Figure 5. In Figure 5a, the proposed ADE method takes amodal masks and masked RGB images as inputs, while in Figure 5b, the depth inpainting techniques take amodal masks and masked depth images as inputs. The quality of the output amodal depth is evaluated in two areas. The restored depth in the masked region is evaluated using the root mean square error (RMSE). The RMSE for the mask region will be referred to as M-RMSE. The depth for the entire area is assessed for accuracy under the threshold in [48] and the RMSE.

4) EVALUATION METRICS ON 3D RECONSTRUCTION

To evaluate the 3D reconstruction performance, a comparison was made with SG3N, a subset of the panoptic 3D reconstruction model [37]. We compared the output of the original SG3N, which takes only a single depth image as input, to that of the proposed ADE, which is given three amodal depth images as input, in terms of both the mesh and occupancy metrics. Standard metrics commonly used in point cloud and mesh studies, such as intersection over union (IoU), chamfer distance (CD), normal consistency (NC), and F score (FS) [49], were also employed. IoU compared SG3N's results to GT voxel data at a resolution of 256^3. CD, NC, and FS were calculated by sampling 40,000 random points from the mesh-formed results and comparing this outcome to the GT.

B. EXPERIMENTAL RESULTS

1) QUANTITATIVE RESULTS

Table 1 presents the quantitative results of the amodal depth outputs. Errors are measured by comparing the outputs of PCNet-C_D, VeI, 3D-Photo, TFill$_D$ [16], [17], [25], and the proposed AMD+ADE with the GT from the Amodal-3D-FRONT dataset. The best values in each metric are highlighted in bold. Notably, the proposed method demonstrates significantly lower M-RMSE values when measuring masked

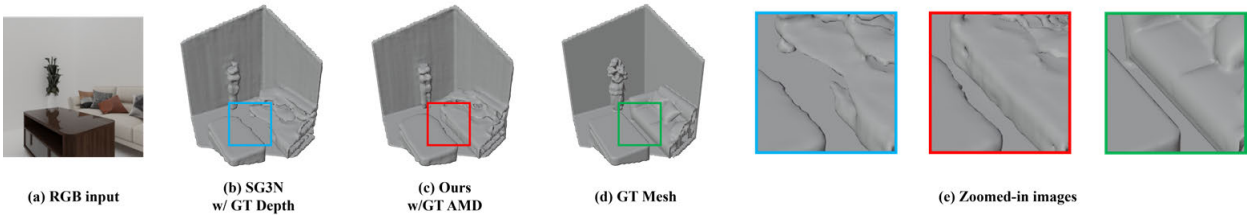


FIGURE 6. Reconstruction quality depending on the provided depth information. This illustrates that the presence or absence of depth information for occluded regions significantly impacts the reconstruction results.

TABLE 1. Quantitative evaluation of amodal depth.

	M-RMSE↓	δ↑	δ^2↑	δ^3↑	RMSE↓
PCNet-C$_D$ [25]	0.3975	0.9318	0.9812	0.9925	0.3104
VeI [16]	0.3002	0.9460	0.9886	0.9979	0.2771
3D-photo [17]	0.2782	0.9607	0.9947	0.9992	0.2458
TFill$_D$ [47]	0.2423	0.9730	0.9963	0.9994	0.2095
Ours	**0.1855**	**0.9841**	**0.9968**	**0.9995**	**0.1534**

TABLE 2. Quantitative evaluation of 3D mesh.

	CD↓	NC↑	FS↑	IoU↑
SG3N [37]	0.1830	0.7980	0.3407	0.3916
Ours	**0.1807**	**0.8058**	**0.3516**	**0.3921**

regions, even 0.0568 lower than TFill$_D$, which has the second lowest M-RMSE. Regarding the overall amodal depth quality, compared to TFill$_D$, the proposed method exhibits a higher δ, δ^2, δ^3 and a lower RMSE. When considering computational complexity, achieving a completely fair comparison proves challenging due to the differing structures of each model. However, assuming similar levels of network models, the proposed method holds advantages. In the proposed structure depicted in Figure 5a, amodal depth can be generated from input images and masks using a single network. On the other hand, as shown in Figure 5b, in most cases, inpainting-based comparative models generate depth through two stages: a depth estimation network and an inpainting network. Simplifying and focusing the network structure, as with ADE, opens up opportunities to design more sophisticated networks, ultimately enhancing output quality.

In Table 2, errors are measured by comparing the 3D reconstruction outputs of the original SG3N, taking a single depth image, and the proposed system with the GT from the Amodal-3D-FRONT dataset. The best values for each metric are shown in bold. Compared to SG3N [37], the proposed system achieved a lower CD by 0.0023. Additionally, it achieved higher NC, FS and IoU values by 0.0078, 0.0109 and 0.0005, respectively. This paper takes advantage of this fact by proposing a model that can iteratively generate amodal depth images from a single RGB image input. By doing so, we demonstrate that it is possible to enhance the performance of 3D reconstruction in the more generalized scenario of a single RGB image input. Unfortunately, the quantitative improvement of the mesh generated by the proposed method is not substantial. As can be seen in the qualitative results of the following subsection, the occluded areas in the 3D mesh constitute only a very small portion of the whole. However, such small differences can have a significant impact on the visual quality and applications utilizing the mesh.

2) QUALITATIVE RESULTS

Figure 7 presents the qualitative results of the proposed ADE. From left to right, Figures 7 (a) and (b) depict the inputs of ADE, which are the masked RGB and amodal mask, respectively. Figures 7 (c) to (f) correspondingly display the depth inpainting outcomes of PCNet-C$_D$ [25], VeI [16], 3D-Photo [17], and TFill$_D$ [47]. Figure 7 (g) presents the output of ADE, whereas Figure 7 (h) is the GT from the Amodal-3D-FRONT dataset. In the results shown in Figures 7 (c), (d), and (e), efforts to restore occlusions are evident, but the distinct appearance of the masked occluder is observed. In Figure 7 (d), relatively good inpainting is achieved for small areas such as chair legs. In Figure 7 (e), there are noticeable pixel flickering artifacts and remnants at the boundaries of the amodal mask. Figure 7 (f) predicts the depth relatively accurately across the entire image area. However, it produces a globally blurry result by inferring the obscured regions through global attention, where original pixel values do not exist. The proposed ADE presents distinct boundaries in the amodal mask and predicts depths for occluded regions closely to the GT without blur artifacts. Owing to the amodal mask, the proposed method accurately restores depths for thin areas such as table legs and chair legs.

The mesh output of the 3D reconstruction is shown in Figure 8. Figure 8 (a) is the RGB input, while Figures 8 (b), (c), and (d) show the outputs of SG3N [37] and the proposed model, as well as the GT Mesh, respectively. The areas of occlusion are highlighted with rectangles, and zoomed-in images are provided in Figure 8 (e). Examining the occlusion areas of the chair and bed in the first row, SG3N leaves the area behind the chair obscured by the bed, while the proposed method successfully restores the side of the bed in its output. In the second row, the table is occluding the sofa. SG3N fails to generate the complete lower part of the sofa, whereas the proposed method accurately reconstructs the shape of the sofa, resembling the GT. Similarly, in the third to sixth rows with various occlusion cases, the proposed model achieves results that are close to the GT.

To evaluate the performance of the proposed structure across various scenarios, additional experiments are

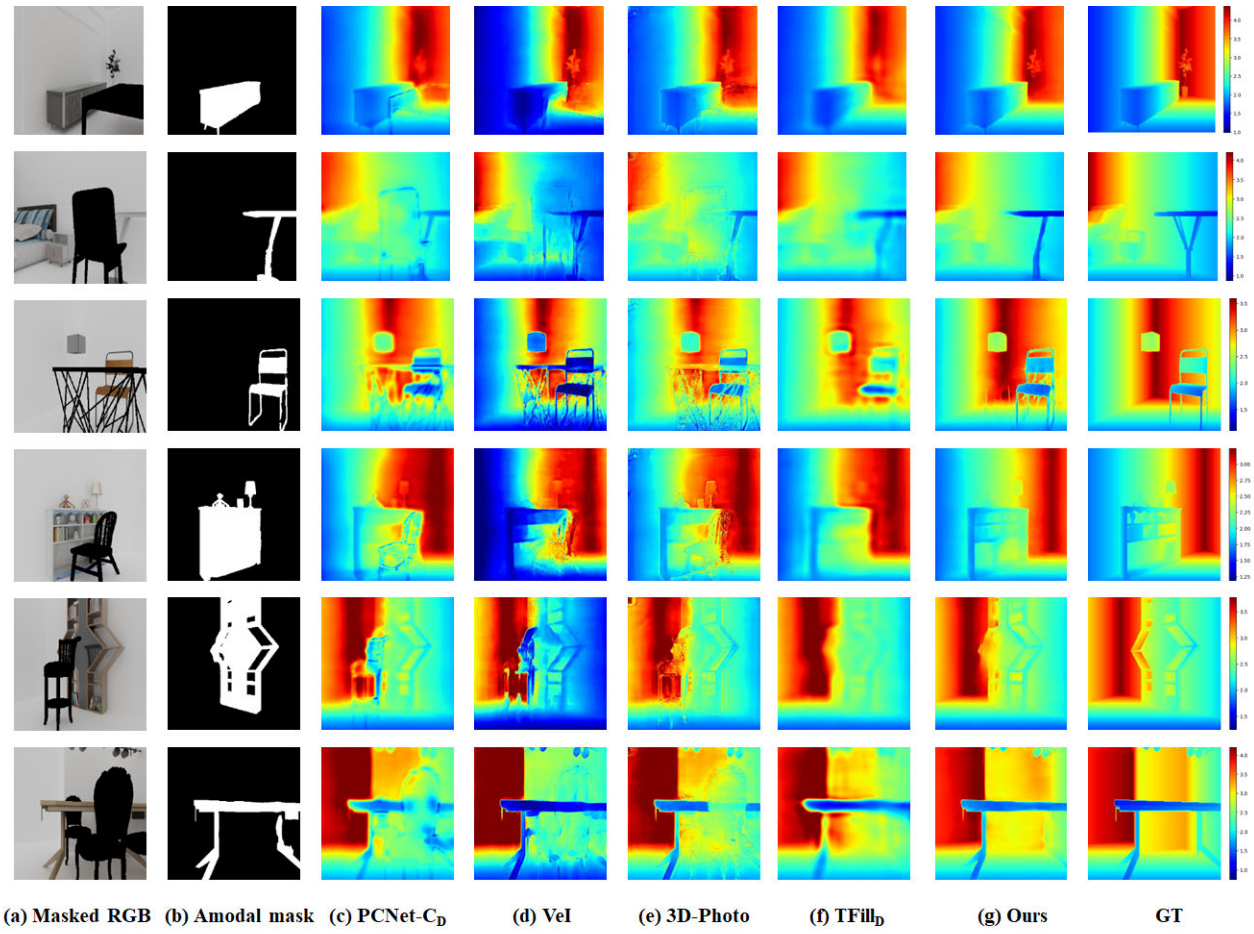


FIGURE 7. Qualitative performance comparison of amodal depth.

conducted using the Matterport3D dataset [44]. Unlike the Front-3D dataset, which provides separate annotations for rooms, walls, floors, and objects, Matterport3D provides scene-level annotations. Therefore, it posed challenges in simulating occlusion scenarios and generating 3D GT. To conduct experiments, we utilized the annotation information of the Matterport3D dataset to create occlusion situations by replicating objects. Additionally, since directly comparing the proposed system's Stage 1 results with GT 3D was challenging, we compared the performance by projecting the inferred depth and Matterport3D's GT depth using camera parameters for 3D projection. Figure 9 displays the results of 3D projection from three amodal depths extracted from RGB images when used the Matterport3D dataset. Experimental results confirm that the 3D projection results generated from depths containing occluded areas through our model can effectively reconstruct 3D in occlusion areas, unlike the projection results using a just single depth.

3) ABLATION STUDY

The effects of the modal mask, amodal mask, and the SPADE fusion block on the accuracy of amodal depth estimation in the proposed model were analyzed. In all cases, the experiments were conducted using the same loss function and training parameters. Comparative experiments were done for different scenarios: when no mask input was provided (denoted by NMsk), when the modal mask was used as input (denoted by MMsk), when the amodal mask was used (denoted by AMsk), and when both the amodal mask and the SPADE fusion block were utilized (denoted by AMsk and SFB). In Table 3, the performance was evaluated using M-RMSE, δ, δ^2, δ^3, and RMSE, similar to Table 1, under various conditions. When no mask was used, the M-RMSE and RMSE values were highest, and δ, δ^2, and δ^3 were lowest, indicating poor amodal depth estimation. As the model mask and amodal mask were introduced, a gradual improvement in the performance was observed. The combination of the

TABLE 3. Ablation study on the quantitative quality of amodal depth.

	M-RMSE↓	δ↑	δ^2↑	δ^3↑	RMSE↓
NMsk	0.6746	0.8584	0.9121	0.9590	0.4966
+MMsk	0.1932	0.9844	**0.9975**	**0.9997**	0.1708
+AMsk	0.1929	0.9812	0.9961	0.9992	0.1635
+AMsk+SFB	**0.1855**	**0.9841**	0.9968	0.9995	**0.1534**

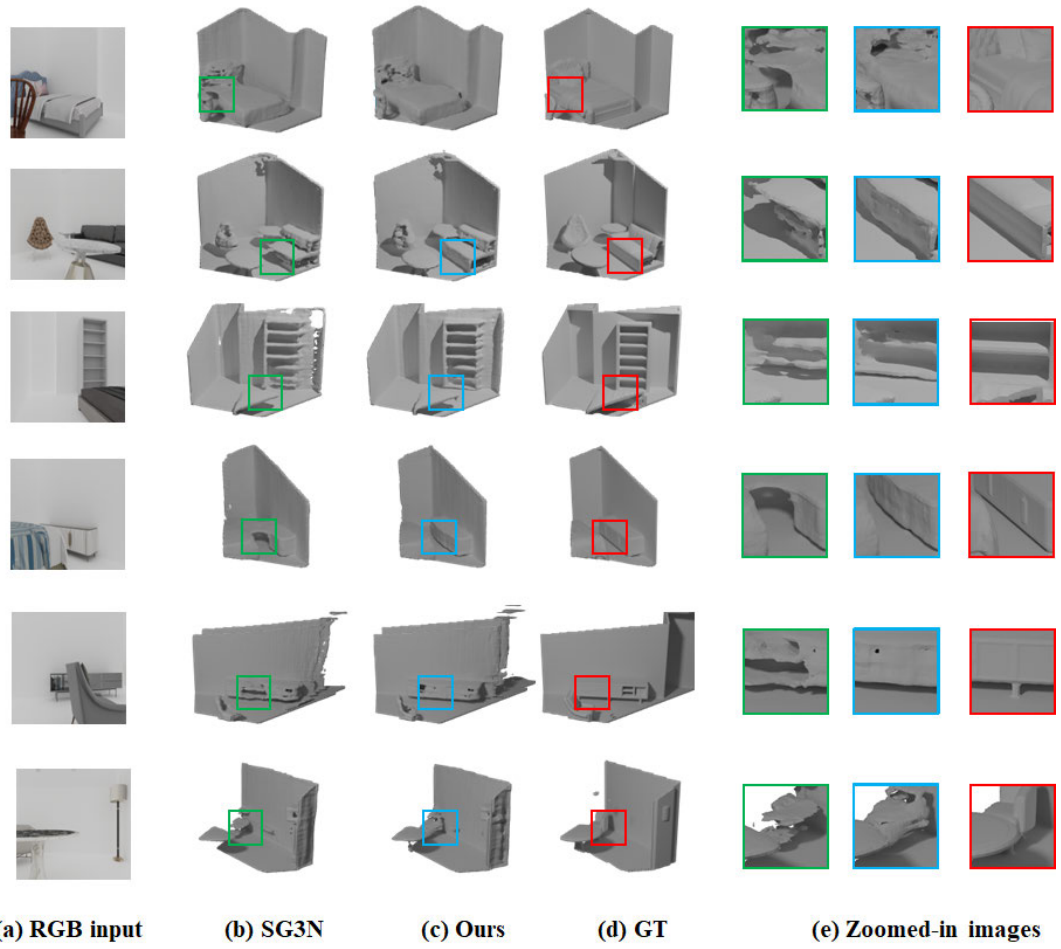


FIGURE 8. Qualitative performance comparison of 3D reconstruction using Front-3D dataset.

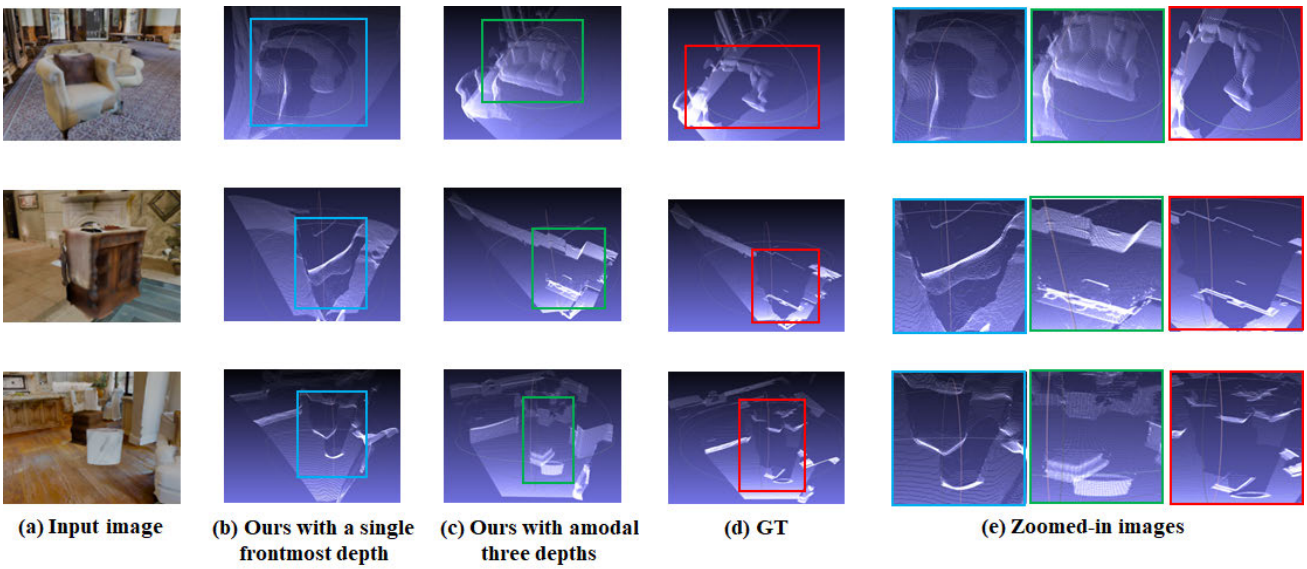


FIGURE 9. Qualitative performance comparison of 3D reconstruction using Matterport3D dataset.

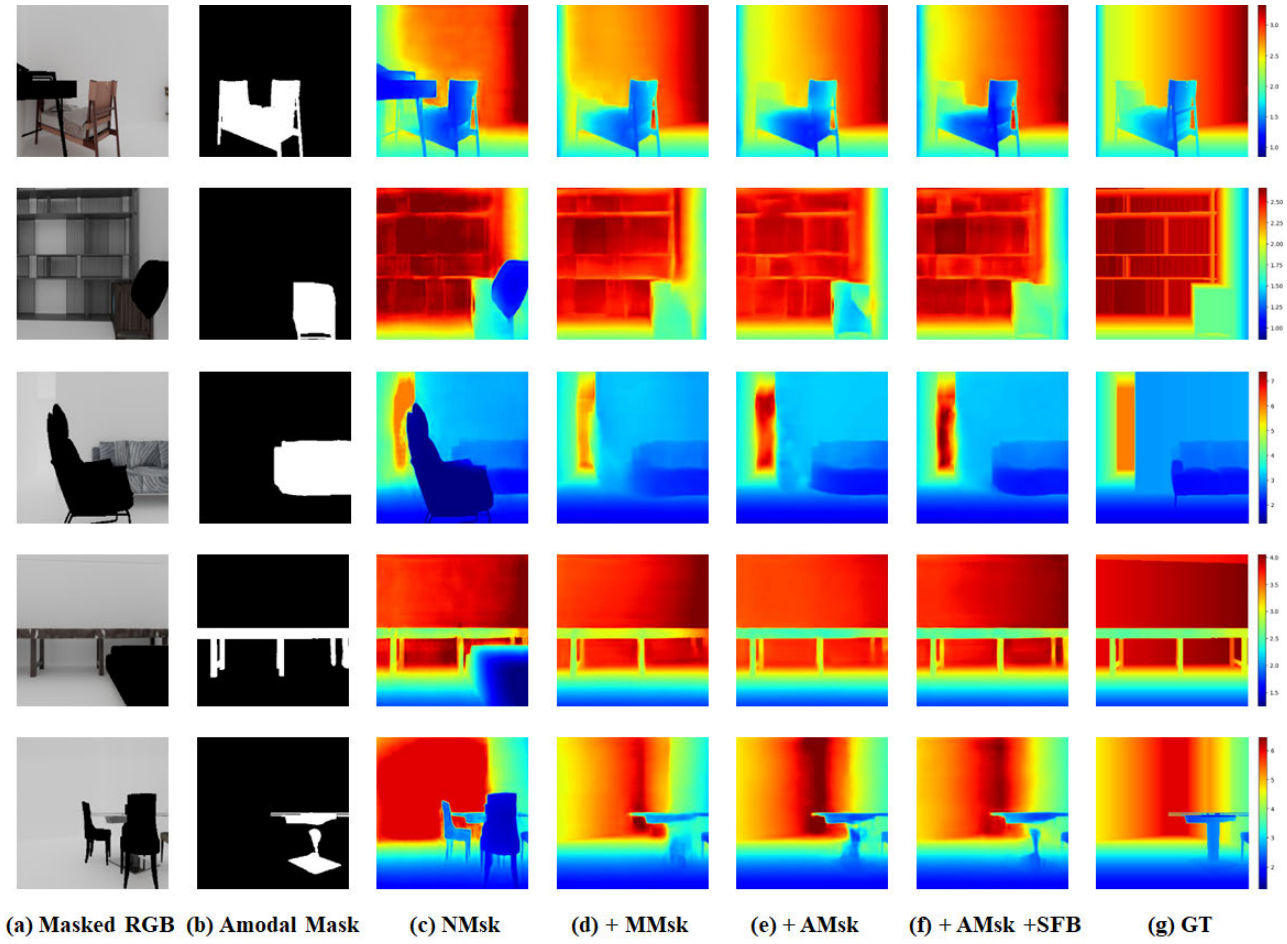


FIGURE 10. Ablation study on the qualitative quality of amodal depth.

amodal mask and the SPADE fusion block yielded the lowest M-RMSE and RMSE values as well as the highest δ value. When considering larger error tolerance ranges such as δ^2 and δ^3, the performance across different conditions becomes somewhat comparable.

Figure 10 qualitatively illustrates the effects of modal masks, amodal masks, and the SPADE fusion block on the amodal depth estimation capabilities. Figures 10 (a) and (b) depict the masked RGB input and the amodal mask used as input to ADE, respectively. Figures 10 (c), (d), and (e) correspond to the cases of No Mask, +MMask, and +AMsk, respectively. Figure 10 (f) presents the outcome when the proposed +AMask+SFB approach is applied, whereas the GT is shown in Figure 10 (g). For the chair occluded by the table in the first row, examining the images from the top of the row, we observe the following: When no mask is provided, the depth of the occluder (table) is directly output, as seen in the topmost image. When only the modal mask is given, the depth of the upper part of the chair, occluded by the table, is not appropriately predicted. In Figure 10 (e), which uses the amodal mask of ADE, we find that the depth of the occluded upper part of the chair is well generated. However, for the shelf in front of the bookshelf in the second row, there are instances where the boundary of the mask is observed in the depth, and as shown in the fourth and fifth rows, inference for thin objects such as table legs is incomplete. In Figure 10 (f), with the full model with the amodal mask and the SPADE fusion block, these problems are well addressed, resulting in predictions that are close to the GT in all cases.

V. CONCLUSION

This paper addresses occlusion challenges in 3D reconstruction using depth estimated from a single image. We propose a technique that introduces an amodal depth prior for 3D reconstruction, which is the first of its kind. By designing a state-of-the-art model to estimate amodal depth from a single image, we validate its capability in enhancing the quality of 3D meshes. Experimental results demonstrate superior performance compared to conventional depth inpainting networks, indicating the potential to enhance occluded region restoration in 3D reconstruction. However, the current approach assumes knowledge of occlusion relationships among objects within the image by providing the modal mask obtained from instance segmentation as input.

Additionally, it lacks generalization across diverse datasets. Future research will focus on outputting multiple amodal depths simultaneously to improve consistency and minimize manual intervention, aiming for end-to-end system-level integration and automation. Expansion to outdoor and real scenes is also necessary.

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